

MTH 553: Lecture # 37

Cliff Sun

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Lecture Span

1. Similarity Method

Similarity Method

A lot of PDEs have similarity solutions defined as

$$u(x, t) = t^\alpha w(t^\beta x) \quad (1)$$

For some constants α, β . For instance, for the one dimensional heat equation

$$u_t = u_{xx} \quad (2)$$

Then we can substitute this similarity solution into this PDE to obtain

$$u_t = t^{\alpha-1} (\alpha w(t^\beta x) + \beta(t^\beta x) w'(t^\beta x)) \quad (3)$$

$$u_{xx} = t^{\alpha+2\beta} w''(t^\beta x) \quad (4)$$

To get the t terms to disappear, let $\alpha + 2\beta = \alpha - 1$ implies that

$$\beta = -\frac{1}{2} \quad (5)$$

Then, we can obtain an equation in terms of w ,

$$w''(y) = \alpha w(y) + \beta y w'(y) \quad (6)$$

$$w''(y) = \alpha w(y) - \frac{1}{2} y w'(y) \quad (7)$$

Where $y = t^\beta x \in \mathbb{R}$. In essence, we have reduced the PDE into an ODE. Then, we can solve this ODE for w . Let $\alpha = \beta$ (note, this choice permits conservation of heat). Therefore, $\alpha = \beta = -1/2$. Then,

$$\int_{-\infty}^{\infty} t^\beta w(t^\beta x) dx = \int_{-\infty}^{\infty} w(y) dy \quad (8)$$

Note that this is independent of t . On the LHS, this is the total heat (since you are integrating the heat equation over x). And this doesn't depend on t which means that $\alpha = \beta = -1/2$ permits heat conservation. We solve

$$w'' = -\frac{1}{2} (w + y w') \quad (9)$$

$$w'' = -\frac{1}{2} (y w')' \quad (10)$$

$$w' = -\frac{1}{2} y w + c_1, \quad \text{choose } c_1 = 0 \quad (11)$$

$$w(y) = C \exp\left(-\frac{y^2}{4}\right) \quad (12)$$

Gaussian equation. Therefore,

$$u(x, t) = t^{-1/2} w(t^{-1/2} x) \quad (13)$$

$$= \frac{C}{(4\pi t)^{1/2}} \exp(-x^2/4t) \quad (14)$$

Remarks

1. The choice of β was forced to eliminate the leading powers of t
2. Other choices of α are mathematically possible. Although not physically possible.

Traveling wave for a non-linear PDE

This is called the Korteweg-de Vries equation (KdV).

$$u_t + 6uu_x + u_{xxx} = 0 \quad (15)$$

Note that this equation is non-linear and third order. Interpretation: u is the height of water or electrical wave in one dimension. Let's try a traveling wave.

$$u(x, t) = v(x - ct) \quad (16)$$

We then obtain

$$-cv'(y) + 6v(y)v'(y) + v'''(y) = 0 \quad (17)$$

Where $y = x - ct$. We can integrate everything

$$-cv + 3v^2 + v'' = a \quad (18)$$

Where a is a constant. Multiply by $2v'$ and integrate

$$(v')^2 = cv^2 - 2v^3 + 2av + b \quad (19)$$

Where b is a constant. This is a non-linear ODE that is first order. Suppose v, v', v'' tend to 0 as $y \rightarrow \pm\infty$. This is a solitary wave. Thus, $b = 0$ and $a = 0$. Therefore, for a solitary wave, we obtain

$$v' = \pm v\sqrt{c - 2v} \quad (20)$$

This is a separable non-linear ODE. Change variable, let

$$v(y) = \frac{c}{2} \operatorname{sech}^2(\theta(y))$$

Then we obtain:

$$\pm v\sqrt{c - 2v} = \pm \frac{c}{2} \operatorname{sech}^2(\theta) \sqrt{c - c \operatorname{sech}^2 \theta} \quad (21)$$

$$= \pm \frac{c^{3/2}}{2} \operatorname{sech}^2 \theta \tanh(\theta) \quad (22)$$

$$v'(y) = -c \operatorname{sech}^2 \theta \tanh \theta \quad (23)$$

Thus

$$\frac{d\theta}{dy} = \pm \frac{\sqrt{c}}{2} \implies \theta = \pm \frac{\sqrt{c}}{2} (y - \phi) \quad (24)$$

Thus,

$$\boxed{v(y) = \frac{c}{2} \operatorname{sech}^2 \left(\frac{\sqrt{c}}{2} [y - \phi] \right)} \quad (25)$$

This is our solitary wave with height $c/2$ with speed c . They are related! In particular, tall waves travel faster. When tall soliton wave overtakes small wave, then the tall wave gets a bump forwards and the small wave gets pushed back. But they keep the same shape for some reason.

A traveling wave is also a similarity solution in logarithmic coordinates.

$$v(x - ct) = v \left(\log \frac{\xi}{\tau^c} \right) \quad (26)$$

Where $\xi = \exp(x)$ and $\tau = e^t$.